

WEEK 4 – ASSIGNMENT

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Task 1:

Explore Azure Portal
Create a Resource Group

The screenshot shows the Microsoft Azure Activity log interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information (jain1heeral@gmail.com). The main header displays 'Home' and 'Activity log'. Below the header, there are filters for 'Activity', 'Edit columns', 'Refresh', 'Export Activity Logs', 'Download as CSV', 'Insights', 'Feedback', 'Pin current filters', and 'Reset filters'. A search bar is present with the text 'Looking for Log Analytics? In Log Analytics you can search for performance, diagnostics, health logs, and more. Visit Log Analytics'. Below the search bar, there are filters for 'Management Group : None', 'Subscription : Azure for Students', 'Event severity : All', 'Timespan : Last 6 hours', and 'Event initiated by : jain1heeral@gmail.com'. A table shows 4 items with columns: Operation name, Status, Time, Time stamp, Subscription, and Event initiated by. The table contains four rows of activity logs, all showing 'Succeeded' status.

Operation name	Status	Time	Time stamp	Subscription	Event initiated by
> Update resource group	Succeeded	4 minutes a...	Sat Jun 13 2026 23:40:55 G...	Azure for Students	jain1heeral@gmail.com
> Update resource group	Succeeded	4 minutes a...	Sat Jun 13 2026 23:40:54 G...	Azure for Students	jain1heeral@gmail.com
> Update resource group	Succeeded	4 minutes a...	Sat Jun 13 2026 23:40:54 G...	Azure for Students	jain1heeral@gmail.com
Create subscription	Succeeded	37 minutes ...	Sat Jun 13 2026 23:08:34 G...	Azure for Students	jain1heeral@gmail.com

Task 2: Storage Setup

- Create a Storage Account

The screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and user information (jain1heeral@gmail.com). The main header displays 'Home' and 'internshipadfstorage_1781416700880 | Overview'. Below the header, there are buttons for 'Delete', 'Cancel', 'Redeploy', 'Download', and 'Refresh'. A large green checkmark indicates 'Your deployment is complete'. Below this, there is a table showing deployment details with columns: Resource, Type, Status, and Ope. The table contains three rows, all showing 'OK' status. To the right of the table, there are sections for 'Cost Management', 'Microsoft Defender for Cloud', and 'Free Microsoft tutorials'. A 'Notifications' panel on the right shows a 'Deployment succeeded' message with a 'Go to resource' button.

Resource	Type	Status	Ope
internshipadfstorage/dei	Microsoft.Storage/stor...	OK	Ope
internshipadfstorage/dei	Microsoft.Storage/stor...	OK	Ope
internshipadfstorage	Microsoft.Storage/stor...	OK	Ope

- Create a Blob Container
- Upload a CSV file

Home > heeral_internship_adf_pipeline > internshipadfstorage | Containers

data
Container

Search

+ Add Directory Upload Change access level Refresh Delete

Overview
Diagnose and solve problems
Access Control (IAM)
Settings

data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive) Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type
Sample - S...	14/6/2026, 10:57:53 PM	Hot (Inferred)	Block blob

Notifications

More events in the activity log → Dismiss all

- Successfully uploaded blob(s)
Successfully uploaded 1 blob(s).
a minute ago
- Successfully created storage container
Successfully created storage container 'data'.
5 minutes ago

Task 3: ADF Basics

- Create Azure Data Factory

Home

Microsoft.DataFactory-20260614230850 | Overview
Deployment

Search Delete Cancel Redeploy Download Refresh

Overview
Inputs
Outputs
Template

Your deployment is complete

Deployment name : Microsoft.DataFactory-20260614230850
Subscription : Azure for Students
Resource group : heeral_internship_adf_pipeline
Start time : 14/6/2026, 11:11:22 PM
Correlation ID : dafd343b-d57f-419c-9802-8e02fafb87d9

Deployment details

Next steps

Go to resource

Give feedback
Tell us about your experience with deployment

Notifications

More events in the activity log → Dismiss all

- Deployment succeeded
Deployment 'Microsoft.DataFactory-20260614230850' to resource group 'heeral_internship_adf_pipeline' was successful.
Pin to dashboard Go to resource
a few seconds ago
- Successfully uploaded blob(s)
Successfully uploaded 1 blob(s).
14 minutes ago
- Successfully created storage container
Successfully created storage container 'data'.
18 minutes ago

- Explore ADF UI (Author, Monitor, Manage)

Microsoft Azure | Data Factory | info

Plan ahead with Fabric Data integration features are being delivered in Fabric first. Explore what's available and plan your migration to Fabric. [Start exploring](#)

Azure Data Factory allows you to configure a Git repository with either Azure DevOps or GitHub. Git is a version control system that allows for easier change tracking and collaboration. [Learn more](#)

[Set up code repository](#)

Data factory info

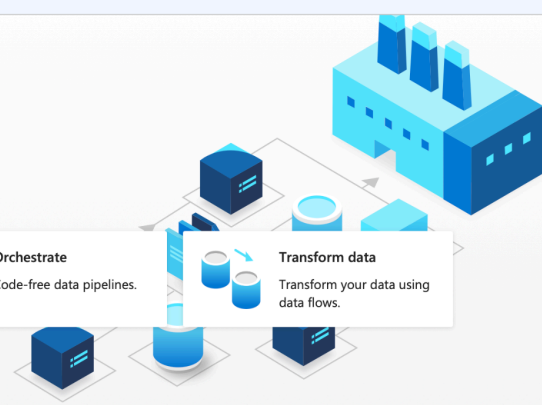
[New](#)

Ingest
Copy data at scale once or on a schedule.

Orchestrate
Code-free data pipelines.

Transform data
Transform your data using data flows.

Configure SSIS
Manage & run your SSIS packages in the cloud.



Microsoft Azure | Data Factory | info

Validate all Publish all

[Migrate to Fabric \(Preview\)](#)

Factory Resources

Filter resources by name

Pipelines	0
Change Data Capture (preview)	0
Datasets	0
Data flows	0
Power Query	0



Select an item
Use the resource explorer to select or create a new item

Microsoft Azure | Data Factory | info

Pipeline runs

Triggered Debug Rerun Cancel options Refresh Edit columns List Gantt

Filter by run ID or name Chennai, Kolkata, Mu... : Last 24 hours

Pipeline name : All Status : All Runs : Latest runs Copy filters Export to CSV

Triggered by : All Add filter

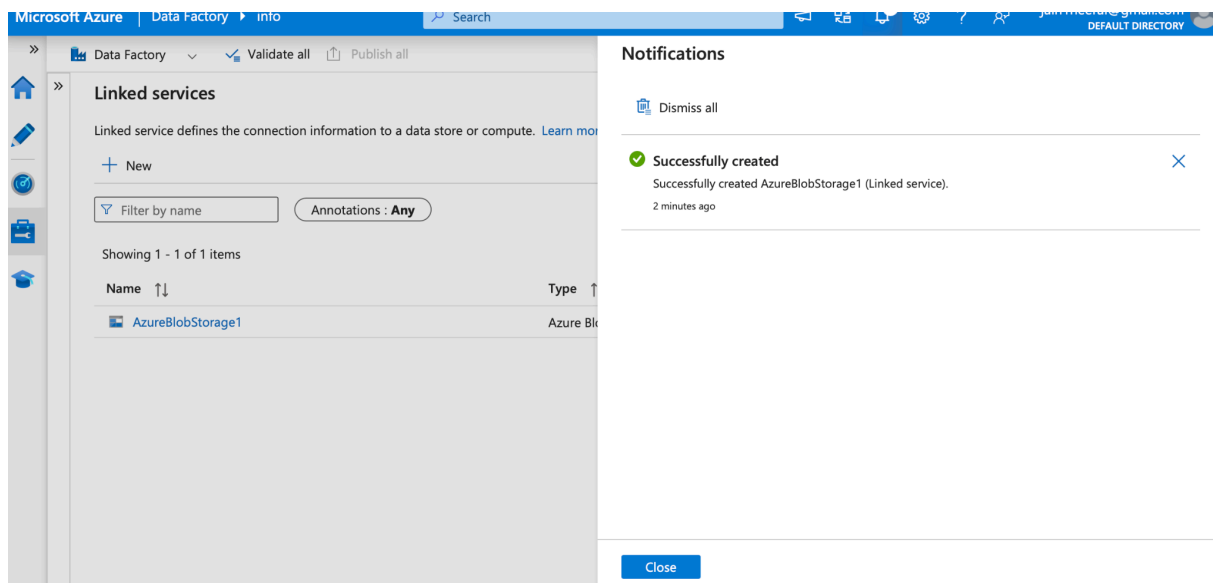
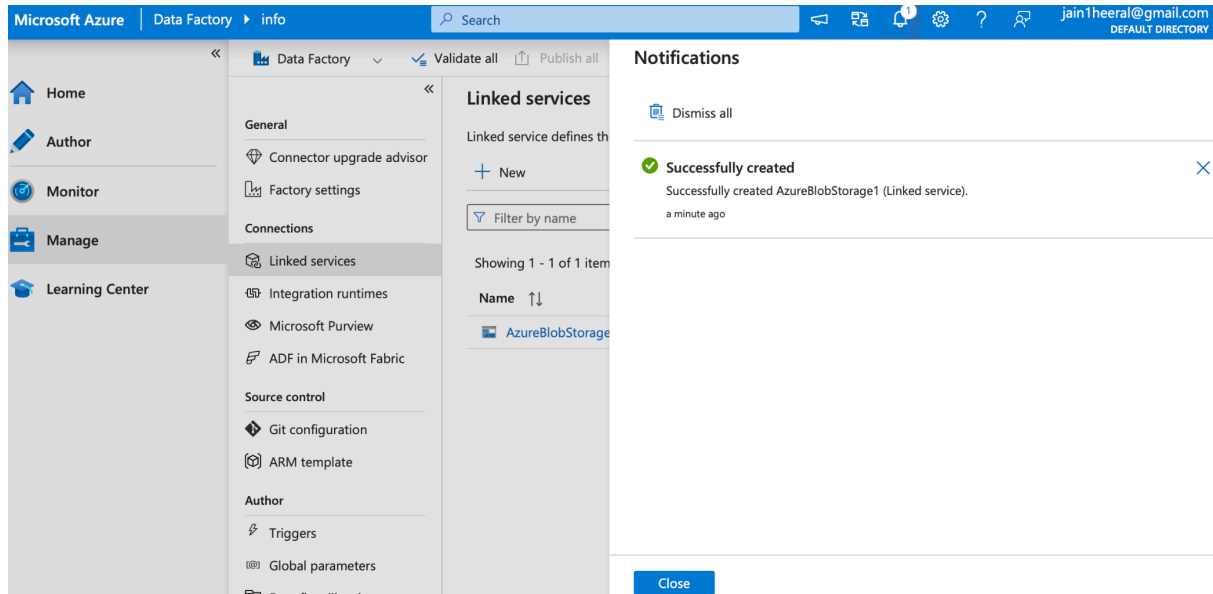
Showing 0 - 0 items Last refreshed 0 minutes ago

Pipeline name	Run start	Run end	Duration
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No results to show
If you expected to see results, try changing your filters.

- Create Linked Service (Blob Storage)



- Create datasets (source + destination)

Microsoft Azure | Data Factory | info

Search

Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 1
 - input
- Data flows 0
- Power Query 0

input

DelimitedText input

Connection Schema Parameters

Linked service *

Test connection Edit

File path *

Compression type

Column delimiter

Row delimiter

Encoding

Notifications

Dismiss all

- Publishing completed**
Successfully published
a minute ago
- Successfully created**
Successfully created AzureBlobStorage1 (Linked service).
15 minutes ago

Close

Microsoft Azure | Data Factory | info

Search

Validate all Publish all

Factory Resources

Filter resources by name

- Pipelines 0
- Change Data Capture (preview) 0
- Datasets 2
 - output
 - input
- Data flows 0
- Power Query 0

output

DelimitedText output

Connection Schema Parameters

Linked service *

Test connection Edit

File path *

Compression type

Column delimiter

Row delimiter

Encoding

Notifications

Dismiss all

- Publishing completed**
Successfully published
a few seconds ago
- Publishing completed**
Successfully published
3 minutes ago
- Successfully created**
Successfully created AzureBlobStorage1 (Linked service).
17 minutes ago

Close

- Use Get Metadata activity

Microsoft Azure | Data Factory | info

Search

jain1heeral@gmail.com

Data Factory | Validate all | Publish all

Migrate to Fabric (Preview)

input | output | pipeline1

Activities

met

General

Get Metadata

Expand resources pane

Get Metadata1

Parameters Variables Settings Output

Pipeline run ID 75dd921e-6126-483b-b7df-b5918e25bb61 Pipeline status Succeeded

All status

Showing 1 - 1 of 1 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Get Metadata1	Succeeded	Get Metadata	6/14/2026, 11:44:49 PM	10s	AutoResolveIntegrationRuntime (East A

Microsoft Azure | Data Factory | info

Search

jain1heeral@gmail.com

Data Factory | Validate all | Publish all

Migrate to Fabric (Preview)

input | output | pipeline1

Activities

co

Move and transform

Copy data

Azure Data Explorer

Azure Data Explorer C...

Iteration & conditionals

Filter

ForEach

If Condition

Switch

Until

Get Metadata1

Copy data

Parameters Variables Settings Output

Pipeline run ID 0041062

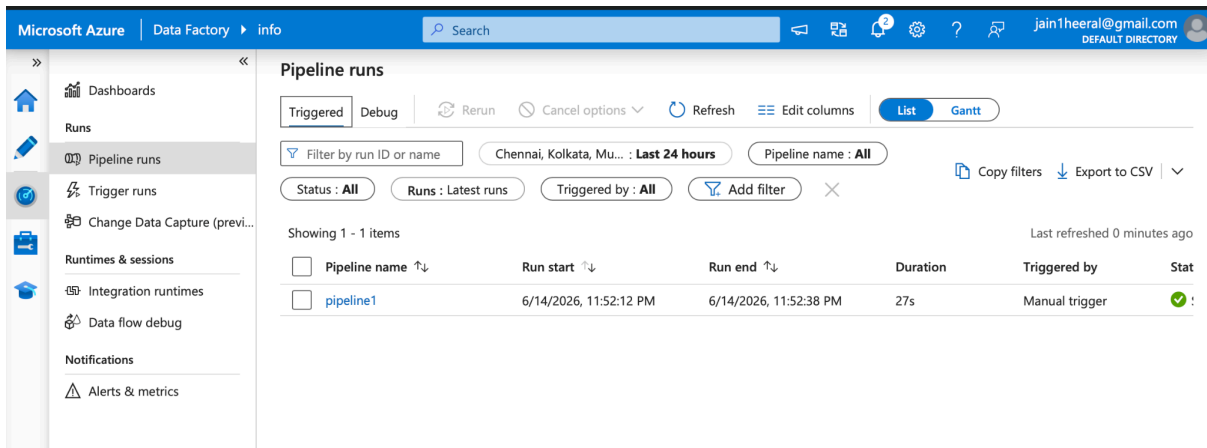
All status

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data	Succeeded	Copy data	6/14/2026, 11:50:22 PM	15s	AutoResolveIntegrationRuntime (Ea
Get Metadata1	Succeeded	Get Metadata	6/14/2026, 11:50:22 PM	13s	AutoResolveIntegrationRuntime (Ea

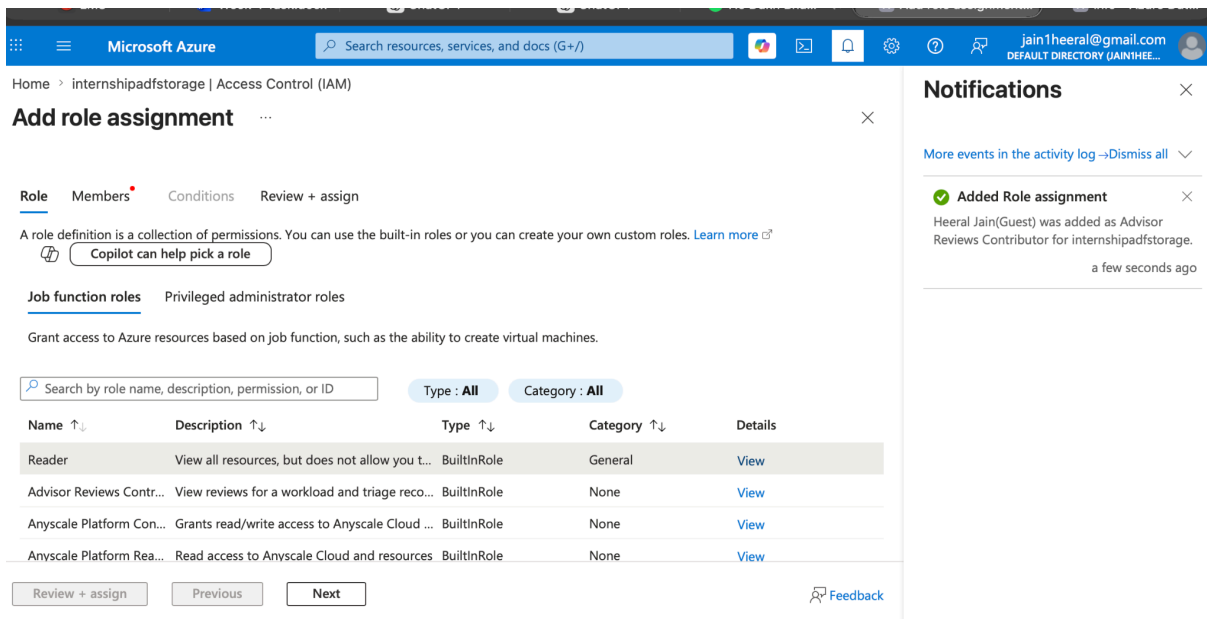
Task 5: Pipeline Execution

- Run pipeline using Debug/Trigger



Task 6: IAM Roles

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage



Build a complete pipeline that reads a CSV file from Blob Storage and processes it using Azure Data Factory.

- Data name(Sample-Store.CSV) is uploaded

Microsoft Azure | Search resources, services, and docs (G+)

Home > heeral_internship_adf_pipeline > internshipadfstorage | Containers

data Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

Name	Last modified	Access tier	Blob type
Sample - S...	14/6/2026, 10:57:53 PM	Hot (Inferred)	Block blob

Notifications

More events in the activity log → Dismiss all

- Successfully uploaded blob(s)
Successfully uploaded 1 blob(s).
a minute ago
- Successfully created storage container
Successfully created storage container 'data'.
5 minutes ago

- The data is being transformed(copy)

Microsoft Azure | Data Factory | info

Data Factory | Validate all | Publish all | Migrate to Fabric (Preview)

input | output | pipeline1

Activities

- Move and transform
 - Copy data
- Azure Data Explorer
 - Azure Data Explorer C...
- Iteration & conditionals
 - Filter
 - ForEach
 - If Condition
 - Switch
 - Until

Get Metadata

Copy data

Parameters Variables Settings Output

run id: 6041062

All status

Showing 1 - 2 of 2 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data	✓ Succeeded	Copy data	6/14/2026, 11:50:22 PM	15s	AutoResolveIntegrationRuntime (Ea
Get Metadata1	✓ Succeeded	Get Metadata	6/14/2026, 11:50:22 PM	13s	AutoResolveIntegrationRuntime (Ea

- Data has been loaded(copied) into the destination named as file **New data**

Microsoft Azure | Search resources, services, and docs (G+)

Home > internshipadfstorage | Containers

data Container

Search

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 2 items

Name	Last modified	Access tier	Blob type	Size	Lease state
New data	15/6/2026, 12:58:12 AM	Hot (Inferred)	Block blob	2.18 MiB	Available
Sample - S...	15/6/2026, 12:54:50 AM	Hot (Inferred)	Block blob	2.18 MiB	Available